Reliability Science

The Chemistry and Physics of Failure – Selected Relationships

Algebraic Rules: Exponents and Logarithms

$a^u > 0$	$\mathrm{Log}_a\, 1 = 0$
$a^0 = 1$	$\mathrm{Log}_a\, a = 1$
$a^u a^v = a^{u+v}$	$\log_a MN = \log_a M + \log_a N$
$a^u/a^v = a^{u-v}$	$\log_a M/N = \log_a M - \log_a N$
$a^{-u} = 1/a^u$	$\log_a M^n = n \log_a M$
$(a^u)^v = a^{uv}$	• The "a" in "$\log_a$" is the **base** of the logarithm.
$a^u b^u = (ab)^u$	• Two common bases are 10 and **e** (Euler's number). e is limit as $t \to 0$ for $(1+t)^{1/t}$ or limit as $n \to 0$ for $(1+n)^n \approx 2.71828$.
$a^{u/v} = (a^u)^{1/v} = (a^{1/v})^u$	

Degradation Analysis - Extrapolation Models

Under normal and even under accelerated stress conditions, reliability testing can take too long for the item under study to fail – and testing to success may also take a prohibitively long time when simulating long field lives (e.g., 15–30 years). An alternative is Degradation Analysis that demonstrates reliability within a relatively short period of time.

Degradation Analysis extrapolates a failure time based on the measurements of degradation or change in performance over time. For an effective degradation analysis, it is necessary to define a level of degradation or performance which constitutes a failure called a **failure threshold**.

Once the failure threshold is established, basic mathematical models are used to extrapolate the performance measurements over time to the point where the failure is expected to occur. Once the extrapolated failure times have been determined, life data analysis should be conducted to model and analyze the demonstrated reliability of the tested items.

Tips: (1) Verify the chosen degradation parameter is affected by the test. This parameter may in reality degrade due to other causes, such as poor quality or be unrelated to the test altogether. (2) The chosen parameter should exhibit a clear degradation trend. It would invalidate the test if the item under study exhibits fluctuations in the parameter value over time.

Common extrapolation models:

- Linear $\boldsymbol{y = ax + b}$
- Power $\boldsymbol{y = ax^b}$
- Exponential $\boldsymbol{y = c^{ax+b}}$ and when c is e (Euler's number), then $Ln\, y = ax + b$
- Log-Log $\boldsymbol{y = e^b x^a}$ via $Ln\, y = aLn\, x + b$

O'Connor, 2012, p. 362

Reliability Science

The Chemistry and Physics of Failure – Selected Relationships

Temperature Related Failures - Arrhenius Model

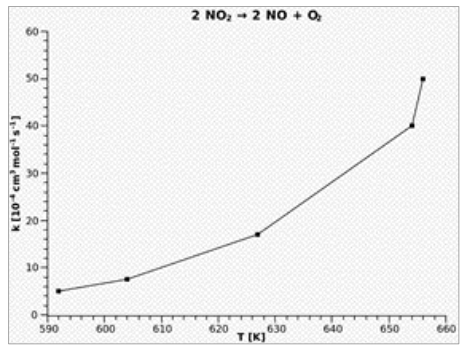
$$A_T = Exp\left\{\frac{E_a}{K_B}\left[\frac{1}{T_{Use}} - \frac{1}{T_{Stress}}\right]\right\}$$

$$Ln(t_f) = C + \frac{E_a}{K_B T}$$

Notation

A_T	=	temperature acceleration factor
T_{Stress}	=	test temperature (°K)
T_{Use}	=	use temperature (°K)
E_a	=	activation energy
K_B	=	8.6173×10^{-5} eV/°K (Boltzmann's constant)
t_f	=	time to failure
C	=	constant

Activation Energy (E_a): 0.7 eV is typically used for integrated circuit (IC) failure mechanisms.



Ebeling, 1997, pp. 327 & 328
Feinberg, 2001, p. 9-4
Wang & Roush, 2000, pp. 33-35

Example (This example uses the Arrhenius model without knowing activation energy, E_a).

Given: An electronic component has a normal operating temperature of 294° K (~21° C, ~70° F). Under stress testing at 430 K (~157 °C, ~314 °F) a Weibull distribution was obtained with a shape parameter β = 1.67 and a scale parameter θ = 254 hours. At 450 K (~177 °C, ~350 °F), a Weibull distribution was obtained with a shape parameter β = 1.77 and θ = 183 hours.

Question: What is the reliability of the component for 1000 hours at normal operating temperature?

Solution:

(1) In the first equation, let $E_a/K_B = B$ and solve for B. The result is: $\boldsymbol{B} = \ln(A_T)/(1/T_1 - 1/T_2)$.

(2) In addition to the first equation, $\boldsymbol{A_T}$ is the ratio of the time-to-failure (TTF) at each stress level. Thus, for $A_T > 1$, then $TTF_1/TTF_2 > 1$. This occurs when $TTF_1 > TTF_2$ and this occurs when $T_1 < T_2$. For this problem, use the two Weibull scale parameters for TTF (Note: At **time θ**, ~63.2% of the failures have occurred; θ is not an average unless the Weibull shape parameter is one). Find A_T when T_1 = 430 K and T_2 = 450 K. That is, $\theta_1/\theta_2 = \frac{254}{183} = 1.39$. Next, use these three values to find B in step 1.

(3) $B = \ln(254/183)/(1/430 - 1/450) = 3171.9$. Now that B is known, find A_T when T_1 = 294 K and T_2 is either 430 K or 450 K, since no θ at 294 K. Using the first equation and T_2 = 450 K …

(4) $A_T = \exp\left[B\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right] = \exp\left[3171.9\left(\frac{1}{294} - \frac{1}{450}\right)\right] = 42.1$. Note: A_T = 30.3 when T_2 = 430 K.

(5) Find the Weibull scale parameter at 294 K. That is, find θ_1 when $A_T = 42.1 = \theta_1/183$ which results in $\theta_1 = 42.1\ x\ 183 = 7704.3$ hours.

(6) Use $\theta_1 = 7704.3$ hours to make the Weibull distribution at 294 K and when t = 1000 and the shape parameter is the average. Solve: $R(t) = \exp\left[-(t/\theta)^{\beta}\right] = \exp\left[-\left(\frac{1000}{7704.3}\right)^{1.72}\right] = 0.9706$.

(7) Thus, based on an accelerated-life test using elevated temperature, this component has a 97% chance in completing a mission with at least 1000 hours when the operating temperature is 294 K (~21 °C, ~70 °F). R(t) via Excel: =1-WEIBULL.DIST(t,β,θ,TRUE) where θ is at the T_{normal} stress level.

Reliability Science

The Chemistry and Physics of Failure – Selected Relationships

Temperature and Humidity Related Failures - Peck Model

$$A_T = Exp\left\{\frac{E_a}{K_B}\left[\frac{1}{T_{Use}} - \frac{1}{T_{Stress}}\right]\right\}$$

$$A_H = \left(\frac{R_{Stress}}{R_{Use}}\right)^m$$

$$A_{TH} = A_T A_H$$

$$Ln(t_f) = C + \frac{E_a}{K_B T} - m \;\; Ln(R)$$

Notation

A_H	=	humidity acceleration factor
A_T	=	temperature acceleration factor
A_{TH}	=	temperature-humidity acceleration factor
R_{Stress}	=	relative humidity of test
R_{Use}	=	nominal use relative humidity
T_{Stress}	=	test temperature
T_{Use}	=	nominal use temperature
m	=	humidity constant
E_a	=	activation energy
t_f	=	time to fail
C	=	constant

Humidity constant (m): Values range between 2.5 and 3.0

Feinberg, 2001, p. 9-6

Temperature Cyclic Related Failures - Coffin Manson Model

$$A_{TC} = \frac{N_{Use}}{N_{Stress}} = \left(\frac{\Delta T_{Stress}}{\Delta T_{Use}}\right)^K$$

$$Ln(N_f) = C - K \;\; Ln(\Delta T)$$

Notation

A_{TC}	=	temperature cycle acceleration factor
N_{Stress}	=	number of cycles tested
N_{Use}	=	equivalent number of field cycles
ΔT_{Stress}	=	temperature cycle test range
ΔT_{Use}	=	nominal daily temperature change in the field
K	=	temperature cycle exponent
N_f	=	number of cycles to failure
C	=	constant

Test items are subjected to a number of cycles of **alternate high and low temperature extremes**. This cyclic stress produced in temperature cycling is related to thermal expansion and contraction undergone in the material.

Temperature Cycle Exponent (K): Values range between 2 to 4 and are design specific.

- 2.5 is commonly used for solder-joint fatigue – and for conservative estimates.
- 4 is commonly used for integrated circuit (IC) failures.

Feinberg, 2001, p. 9-7

Reliability Science

The Chemistry and Physics of Failure – Selected Relationships

Vibration Related Failures - MIL-STD 810E

$$A_V = \frac{T_{Use}}{T_{Stress}} = \left(\frac{W_{Stress}}{W_{Use}}\right)^{M_b}$$

$$\left(\frac{W_{Stress}}{W_{Use}}\right)^{M_b} = \left(\frac{G_{f_{Stress}}}{G_{f_{Use}}}\right)^{2M_b}$$

$$Ln(t_f) = C - M_b \; Ln(W)$$

Notation

A_V = vibration acceleration factor
T_{Stress} = vibration duration
T_{Use} = vibration duration (nominal)
W = random vibration input PSD across the resonance bandwidth (G^2/Hz) W_{Stress} is the PSD test stress and W_{Use} nominal use PSD
G_f = resonant G sinusoid vibration level
M_b = b/2 where b is the fatigue parameter
t_f = time to failure
C = constant

Devices are mounted on a dynamic shaker table and subject to either a random or sinusoidal-type vibration profile.

The traditional classical time-compression model (MIL-STD 810E) is a power law model.

In a random vibration-loading environment, the resonance of the material can dominate the fatigue life--here, maximum vibration amplitudes and stress occur.

However, fatigue failures are not always dominated by the fundamental resonance mode. In practice, many of the stress peaks in the use environment may fall below the fatigue limit of the material, while others will be above the fatigue limit. It follows that an accelerated life test based upon this model should inherently be conservative.

Fatigue Parameter (b): b ≈ 5 is commonly used for electronic boards. A conservative value is about 8 (e.g., Mb = 4). MIL STD810E (514.4-46) recommends b = 8 for random loading.

Feinberg, 2001, p. 9-8

Reliability Science

The Chemistry and Physics of Failure – Selected Relationships

Electromigration Failures - Black equation

$$A_J = \left(\frac{J_{Stress}}{J_{Use}}\right)^n Exp\left\{\frac{E_a}{K_B}\left[\frac{1}{T_{Use}} - \frac{1}{T_{Stress}}\right]\right\}$$

$$Ln(t_f) = C + \frac{E_a}{K_B T} - nLn(J)$$

Notation

A_J	=	electromigration acceleration factor
T_{Stress}	=	test temperature (°K)
T_{Use}	=	use temperature (°K)
E_a	=	activation energy
K_B	=	8.6173×10^{-5} eV/°K (Boltzmann's constant)
J	=	current density
n	=	current density exponent
t_f	=	time to failure
C	=	constant

Electromigration is a failure mechanism caused in a microelectronic conductor exposed to high current densities or a combination of high temperature and current density. The most common failure mode is a conductor open.

Activation Energy (E_a): Values range between 0.5 to 1.1 eV

Current Density Exponent (n): Values range between 2 and 3.3

As lower values of E_a and n are used, the life estimate becomes more conservative.

Feinberg, 2001, p. 9-9

Wearout (Fatigue) Failures via Uniform Cyclic Load - S-N Curve

$$N = cS\hat{}(-m)$$

Notation

N	=	median number of cycles to failure
S	=	magnitude of the cyclic stress
c and m	=	constants determined experimentally

Data Analysis

The log-log plot of the stress S and the number of cycles N to failure is called the **S–N curve**. See next section for an example.

The S–N curve indicates the cycles to failure at any cyclic stress value between the ultimate stress and the fatigue limit (endurance limit).

Log S vs. Log N data tend to fall along a straight line. The values for c and m can be obtained from a least squares analysis.

Fatigue damage within engineering materials is caused when a repeated mechanical stress is applied, the stress being above a limiting value called the **fatigue limit** (endurance limit). Fatigue damage is cumulative, so that repeated fatigue above the fatigue limit will eventually result in failure.

Below the fatigue limit the life is indefinite, but higher stress levels induce cumulative damage, leading ultimately to failure.

Fatigue strength (maximum stress for a specific number of load cycles) is generally less than would be observed under a **static load** (failure due to nearly instantaneous stress).

Certain materials, such as steel, have an infinite life below a specific stress level (endurance limit).

Ebeling, 1997, p. 159
Feinberg, 2001, pp. 14-12
O'Connor, 2012, pp. 208 & 209
Wang & Roush, 2000, p. 29

Reliability Science

The Chemistry and Physics of Failure – Selected Relationships

Wearout (Fatigue) Failures via Randomly Distributed Stresses – Miner's Rule

$$\frac{n_1}{N_1} + \frac{n_2}{N_2} + \cdots + \frac{n_k}{N_k} = \sum_1^k \frac{n_i}{N_i} = 1$$

Notation

n_i = number of cycles at stress level i above the fatigue limit

N_i = median number of cycles to failure at stress level i via the S–N curve

Note: Fatigue life of an item subject to an alternating stress with a mean value of zero is $\sum n_i = N_e$ being **equivalent life**.

Alternate Form

$$\sum_1^k \frac{t_i}{L_i} = 1$$

t_i = amount of time at stress level i

L_i = expected life at stress level i

The S–N curve in the previous section addressed the simplest situation of fatigue life in which a uniform cyclic load is applied. When the stresses are randomly distributed stresses (varying stress), estimate fatigue life using Palmgren-Miner's Law (more often referred as **Miner's rule**).

Miner's rule has the form $\sum \left({}^{n_i}/_{N_i} \right) = 1$ where $\boldsymbol{n_i}$ and $\boldsymbol{N_i}$ are determined from the S-N curve.

Miner's rule applies if cyclic work can be approximated by the number of cycles times an average amount of work per cycle of a certain size. However, if the work is some nonlinear function of the number of cycles, then Miner's rule is invalid.

Ebeling, 1997, p. 330
Feinberg, 2001, pp. 14-8 & 14-9
O'Connor, 2012, pp. 210 - 212

Example 1 (This example is from Miner's original paper in the Journal of Applied Mechanics, 1945)
Given: A pressure vessel operates in two states at 20 cycles per minute. The first state is 60% of the time at a stress change of 1500 psi; the cycles to failure in this state are 1,000,000 cycles. The second state (40% of the time) at a stress change of 2400 psi; the cycles to failure in this state are 215,000.
Question: What is the expected life in hours of this hydraulic pressure unit?

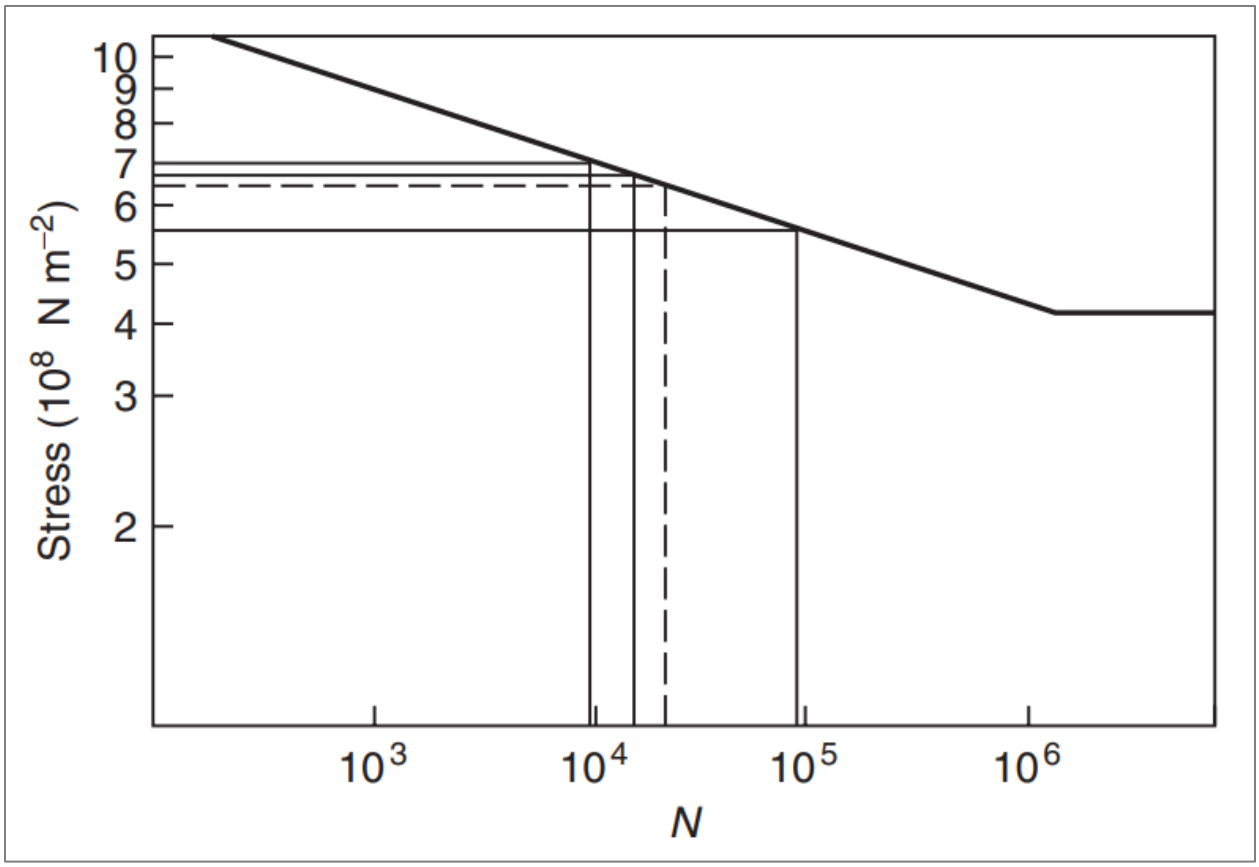
Solution: (1) $n_1 + n_2 = 20$ cycles per minute; $n_1 = 0.6 \times 20 = 12$ cycles per minute and $n_2 = 0.4 \times 20 = 8$ cycles per minute. (2) Using Miner's rule: $\text{Damage} = \frac{12}{1{,}000{,}000} + \frac{8}{215{,}000} = 4.92x10^{-5}$ per minute = 0.00295 per hour. (3) Miner's rule defines a failure as "Damage = 1" at duration N, and the problem determined the damage rate is 0.00295 per hour. Thus, solve for N hours. (4) That is, find N using 1=(N hours/1 failure)*(0.00295 damage/1 hour). N ≈ 339 hours of expected life with mixed stresses.

Example 2 (The S-N curve is a prerequisite analysis and is provided on the next page)

Given: Load data on a component indicate three values exceed the fatigue limit stress of 4.5×10^8 Nm^{-2}. These values occur during operation in the following proportions: (a) 5.5×10^8 Nm^{-2} : 3, (b) 6.5×10^8 Nm^{-2} : 2, and (c) 7.0×10^8 Nm^{-2} : 1. The S-N curve for this material (next page) indicates the cycles to failure at each overstress level are: (a) 5.5×10^8 Nm^{-2} : 9.5×10^4 cycles, (b) 6.5×10^8 Nm^{-2} : 1.5×10^4 cycles, and (c) 7.0×10^8 Nm^{-2} : 0.98×10^4 cycles.

Question: What is the *equivalent* constant dynamic stress?

Solution: (1) Using Miner's rule and an arbitrary constant (C), $3C/(9.5\ x\ 10\text{^}4) + 2C/(1.5\ x\ 10\text{^}4) + 1C/(0.98\ x\ 10\text{^}4\ ) = 1$. (2) Solving for C yields 3746. (3) Using C and the equivalent life relationship, $N_e = 3C + 2C + 1C = 2.25\ x10^4$ cycles. (4) From the S-N curve, when cycles = 2.25×10^4, the equivalent constant dynamic stress is 6.3×10^8 N/m^2 (denoted as the dashed lines in the S-N curve).


Stress (10^8 N m^{-2})
10
9
8
7
6
5
4
3
2
10^3
10^4
10^5
10^6
N